

Администрирование сетевых подсистем

Лабораторная работа №1 — Развёртывание лабораторного стенда с использованием Vagrant

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Цели е задачи работы

Цель лабораторной работы

Овладеть практическими навыками установки операционной системы Rocky Linux на виртуальную машину с помощью инструмента Vagrant и автоматизации процесса развёртывания.

Выполнение лабораторной работы

Подготовка среды

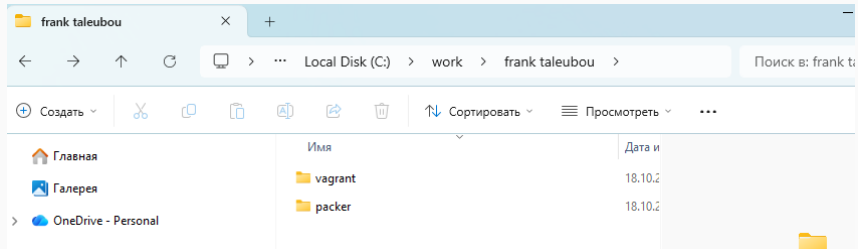


Рис. 1: Подготовка среды VirtualBox и Vagrant

Создание файлов проекта

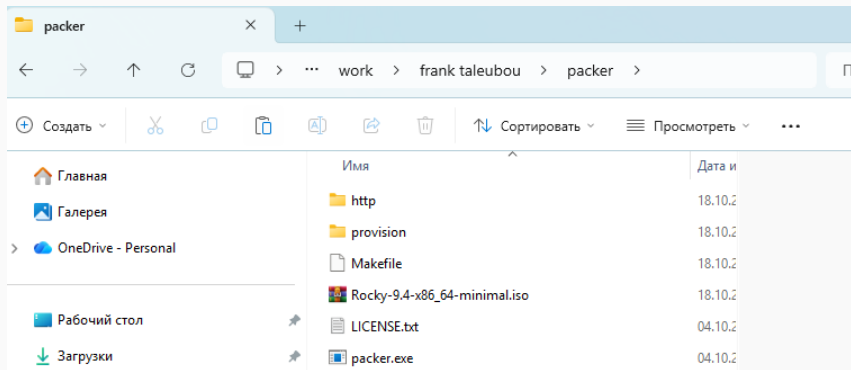


Рис. 2: Скрипт 01-user.sh

Создание файлов проекта

```
1 # -*- mode: ruby -*-
2 # vi: set ft=ruby :
3
4 Vagrant.configure("2") do |config|
5
6   ## Common configuration
7   config.vm.provision "common dummy",
8     type: "shell",
9     preserve_order: true,
10    path: "provision/default/01-dummy.sh"
11
12   config.vm.provision "common hostname",
13     type: "shell",
14     preserve_order: true,
15     run: "always",
16     path: "provision/default/01-hostname.sh"
17
18   config.vm.provision "common user",
19     type: "shell",
20     preserve_order: true,
21     path: "provision/default/01-user.sh"
22
23   ## Server configuration
24   config.vm.define "server", autostart: false do |server|
25     server.vm.box = "rocky9"
26     server.vm.hostname = 'server'
27
28     server.vm.boot_timeout = 1440
29
30     server.ssh.insert_key = false
31     server.ssh.username = 'vagrant'
32     server.ssh.password = 'vagrant'
33
34     server.vm.network :private_network,
35       ip: "192.168.1.1",
36       virtualbox____intnet: true
37
38     server.vm.provision "server dummy",
39       type: "shell",
40       preserve_order: true,
41       path: "provision/server/01-dummy.sh"
42
43     server.vm.provider :virtualbox do |v|
44       v.linked_clone = true
45
46       # Customize the amount of memory on the VM
47       v.memory = 8192
48       v.cpus = 5
49       v.name = "server"
```

Конфигурация Vagrantfile

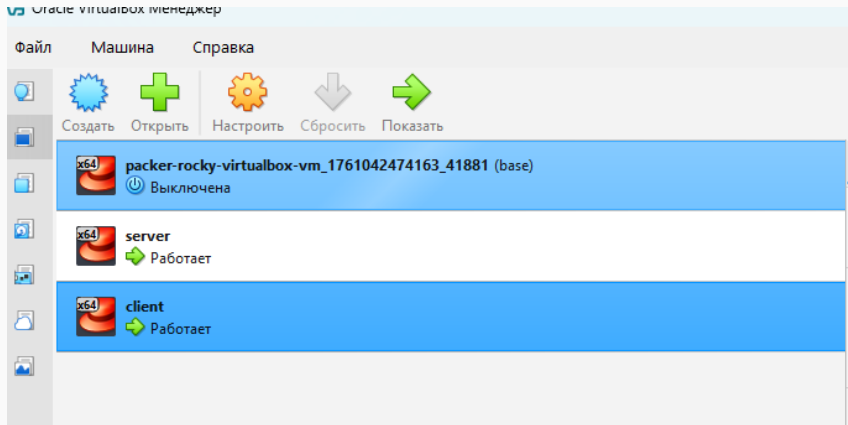


Рис. 4: Файл Vagrantfile с описанием provision

Регистрация box-файла em Vagrant

```
PS C:\work\frank talebou\vagrant> vagrant up server --provision
Bringing machine 'server' up with 'virtualbox' provider...
==> server: This machine used to live in C:/work/franktalebou/user/user/vagrant but it's now at C:/work/frank talebou/
vagrant.
==> server: Depending on your current provider you may need to change the name of
==> server: the machine to run it as a different machine.
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Clearing any previously set forwarded ports...
==> server: Clearing any previously set network interfaces...
==> server: Preparing network interfaces based on configuration...
```

Рис. 5: Добавление box-файла em систему Vagrant

Запуск виртуальной машины Server

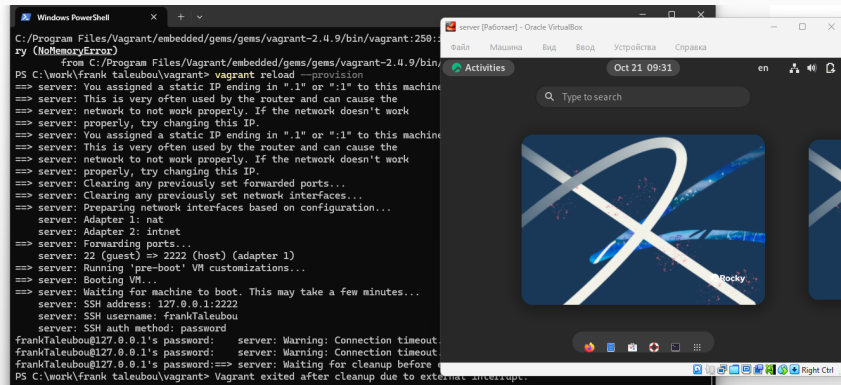


Рис. 6: Запуск виртуальной машины server

Проверка входа em систему

```
PS C:\work\frank taleubou\vagrant> vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering the
==> server: password for you. If you're prompted for a password, please enter
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Tue Oct 21 12:16:58 2025 from 10.0.2.2
[vagrant@server ~]$ su - frank
su: user frank does not exist or the user entry does not contain all the required fields
[vagrant@server ~]$ su - frank
su: user frank does not exist or the user entry does not contain all the required fields
[vagrant@server ~]$ su - frank
su: user frank does not exist or the user entry does not contain all the required fields
[vagrant@server ~]$ su - frank
su: user frank does not exist or the user entry does not contain all the required fields
[vagrant@server ~]$ s
```

Рис. 7: Окно входа em систему Rocky Linux

Проверка имени хоста

```
PS C:\work\frank taleubou\vagrant> vagrant up client --provision
Bringing machine 'client' up with 'virtualbox' provider...
==> client: This machine used to live in C:/work/franktaleubou/user/user/vagrant but it's now at C:/work/frank taleubou/
vagrant.
==> client: Depending on your current provider you may need to change the name of
==> client: the machine to run it as a different machine.
==> client: Clearing any previously set forwarded ports...
==> client: Fixed port collision for 22 => 2222. Now on port 2200.
==> client: Clearing any previously set network interfaces...
==> client: Preparing network interfaces based on configuration...
      client: Adapter 1: nat
      client: Adapter 2: intnet
==> client: Forwarding ports...
      client: 22 (guest) => 2200 (host) (adapter 1)
==> client: Running 'pre-boot' VM customizations...
==> client: Booting VM...
```

Рис. 8: Отображение hostname server.akabrikosov.net

Выводы по проделанной работе

В результате работы был развернут лабораторный стенд на базе VirtualBox и Vagrant.

Были созданы и протестированы сценарии автоматического развёртывания виртуальной машины с использованием скриптов 01-user.sh и 01-hostname.sh.

Проверена корректность настроек пользователя, имени хоста и подключения по SSH.